

Appendix

Proof of Proposition 1. Let $\mu(x) = P(\theta = 1|x)$ be the voter's posterior belief that the incumbent is aligned given the realized policy outcome x . As effort is unobserved, let the voter have conjecture about the incumbent's effort choice, $\hat{a}_\theta$. Formally, posterior beliefs can be expressed as

$$\mu(x) = \frac{\gamma\phi(\sqrt{\zeta}(x - \hat{a}_1))}{\gamma\phi(\sqrt{\zeta}(x - \hat{a}_1)) + (1 - \gamma)\phi(\sqrt{\zeta}(x - \hat{a}_0))}.$$

The voter retains the incumbent iff $\mu(x) \geq \gamma$, which is equivalent to

$$x \geq \frac{\hat{a}_1 + \hat{a}_0}{2}.$$

Given $x = a + \varepsilon$, the incumbent leader survives iff $a + \varepsilon \geq \frac{\hat{a}_1 + \hat{a}_0}{2}$. Since $\varepsilon \sim N(0, \frac{1}{\zeta})$, the incumbent's reelection probability is equal to

$$\pi(a) = \Phi(\sqrt{\zeta}(a - \frac{\hat{a}_1 + \hat{a}_0}{2})).$$

The leader of type θ maximizes

$$\max_a \pi(a)\Psi - \frac{\lambda_\theta}{2}a^2.$$

This leads to the first-order condition

$$-\lambda_\theta a + \sqrt{\zeta}\phi(\sqrt{\zeta}(a - \frac{\hat{a}_1 + \hat{a}_0}{2}))\Psi = 0.$$

Since beliefs are correct in equilibrium, $a_\theta = \hat{a}_\theta = a_\theta^*$, this simplifies to

$$-\lambda_\theta a_\theta^* + \sqrt{\zeta}\phi(\sqrt{\zeta}(\frac{a_1^* - a_0^*}{2}))\Psi = 0.$$

Substituting in $\theta = 1$ and $\theta = 0$ yields the two equations in the proposition.

To show that this solution is a maximum, we ensure that the leader's utility is concave. The second-order condition is

$$-\lambda_\theta - \zeta^{3/2}(a - \frac{\hat{a}_1 + \hat{a}_0}{2})\phi(\sqrt{\zeta}(a - \frac{\hat{a}_1 + \hat{a}_0}{2}))\Psi.$$

Let $\eta = \sqrt{\zeta}(a - \frac{\hat{a}_1 + \hat{a}_0}{2})$ so the second-order condition can be rewritten as

$$-\lambda_\theta - \zeta\eta\phi(\eta)\Psi.$$

The standard normal density tends to zero faster than any polynomial so $\eta\phi(\eta)$ is zero at

$\eta = 0$ and approaches zero as $\eta \rightarrow \pm\infty$. The derivative of $\eta\phi(\eta)$ is $\phi(\eta) - \eta^2\phi(\eta)$ with critical points at $\eta = \pm 1$. Note that if $\eta = 1$ then the problem is globally concave. Hence the relevant constraint is at $\eta = -1$, where $\eta\phi(\eta) = \frac{1}{\sqrt{2\pi e}}$. Hence the leader's utility is concave iff

$$-\lambda_\theta + \frac{\zeta}{\sqrt{2\pi e}}\Psi < 0,$$

or $\zeta < \frac{\lambda_\theta\sqrt{2\pi e}}{\Psi}$. Hence a sufficient condition for both leaders to have concave utility functions is $\zeta < \frac{\lambda_1\sqrt{2\pi e}}{\Psi}$.

Furthermore, this equilibrium is unique because pooling cannot be an equilibrium. Since $\lambda_0 > \lambda_1$, efforts are always distinct unless both leaders were to pool on $a_\theta = 0$. But clearly the aligned leader has incentive to deviate from this strategy as it would increase the odds of reelection. $\square$

Proof of Proposition 2. Proof is analogous to that of Proposition 1 for the derivation of the optimal effort. The only difference is the derivation of the voter's policy cutoff, which is a function of conjectures about the leader's effort $\hat{a}_\theta$ as well as conjectures about the pledges $\hat{p}_\theta$.

Denote $\mu(x, s)$ as the voter's posterior belief about the leader's type having observed IO report s and signal x of the leader's effort. Since the leader's true pledge m and true effort a are unobserved, the voter needs to have conjectures. Let $\hat{a}_\theta$ be the voter's conjecture about leader-type θ 's effort, and let $\hat{p}_\theta = P(m = 1|\theta)$ be the voter's conjecture about the probability that leader-type θ sent pledge $m = 1$ to the IO. Then $\hat{m}_\theta = \hat{p}_\theta\phi(\sqrt{\tau}(s-1)) + (1-\hat{p}_\theta)\phi(\sqrt{\tau}s)$ is the total probability that that IO's report is realized as the value s given voter's conjectures. Then $\mu(x, s)$ can be expressed as

$$\mu(x, s) = \frac{\gamma\phi(\sqrt{\zeta}(x - \hat{a}_1))\hat{m}_1}{\gamma\phi(\sqrt{\zeta}(x - \hat{a}_1))\hat{m}_1 + (1 - \gamma)\phi(\sqrt{\zeta}(x - \hat{a}_0))\hat{m}_0},$$

such that it is optimal to retain the incumbent leader whenever

$$x \geq \frac{\hat{a}_1 + \hat{a}_0}{2} + \frac{\log(\frac{\hat{m}_0}{\hat{m}_1})}{\zeta(\hat{a}_1 - \hat{a}_0)} \equiv \hat{x}(\hat{a}, \hat{p}).$$

It is immediate that whenever $\hat{p}_1 = \hat{p}_0$ then $\hat{x}(\hat{a}, \hat{p}) = \frac{\hat{a}_1 + \hat{a}_0}{2}$, as in the model without the IO. Optimal effort is thus identical to that characterized in Proposition 1.

Suppose $\hat{p}_1 \neq \hat{p}_0$. The first-order condition for leader-type θ 's effort is

$$-\lambda_\theta a + \sqrt{\zeta}\phi(\sqrt{\zeta}(a - \frac{\hat{a}_1 + \hat{a}_0}{2} - \frac{\log(\frac{\hat{m}_0}{\hat{m}_1})}{\zeta(\hat{a}_1 - \hat{a}_0)}))\Psi = 0.$$

Because the report likelihood ratio is monotonic in s when $\hat{p}_1 \neq \hat{p}_0$, the voter's retention cutoff moves monotonically with the IO report. The leader's marginal electoral return, and hence her optimal effort, is single-peaked in this cutoff. It follows that $a_\theta^*(s)$ is continuous

and single-peaked in s . Moreover, at $s = 1/2$, the report is equally likely under either pledge, so it provides no information about the leader's type and $a_\theta^*(1/2) = \tilde{a}_\theta$. Therefore, the set of reports for which $a_\theta^*(s) \geq \tilde{a}_\theta$ is a nonempty, possibly unbounded interval. Denote its endpoints by $\underline{s}_\theta$ and $\bar{s}_\theta$. $\square$

Proof of Proposition 3. The leader maximizes

$$\max_{m \in \{0,1\}} \int_{-\infty}^{\infty} \left[\pi(a_\theta^*(s))\Psi - \frac{\lambda_\theta}{2} a_\theta^*(s)^2 \right] \sqrt{\tau} \phi(\sqrt{\tau}(s - m)) ds.$$

Define $V_\theta(s) = \max_a \pi(a_\theta(s))\Psi - \frac{\lambda_\theta}{2} a_\theta(s)^2$ as leader-type θ 's value function given the optimal effort choice and recall that $p_\theta = P(m = 1|\theta)$. The difference in the leader's utility from choosing $m = 1$ over $m = 0$ is thus

$$\Delta_\theta(p_1, p_0) = \underbrace{\int_{-\infty}^{\infty} V_\theta(s) \sqrt{\tau} \phi(\sqrt{\tau}(s - 1)) ds}_{E[V_\theta(s)|m=1]} - \underbrace{\int_{-\infty}^{\infty} V_\theta(s) \sqrt{\tau} \phi(\sqrt{\tau}s) ds}_{E[V_\theta(s)|m=0]}.$$

It is clear that $m = 1$ is optimal iff $\Delta_\theta(p_1, p_0) > 0$, $m = 0$ is optimal iff $\Delta_\theta(p_1, p_0) < 0$, and the leader is indifferent iff $\Delta_\theta(p_1, p_0) = 0$.

Observe that by the envelope theorem,

$$\frac{dV_\theta(s)}{ds} = -\sqrt{\zeta} \Psi \phi(\sqrt{\zeta}(a_\theta^* - \hat{x}(s; p))) \frac{d\hat{x}(s; p)}{ds},$$

and note that since the distribution of s has the MLRP, $\frac{d\hat{x}(s; p)}{ds} > 0$ iff $p_0 > p_1$ and $\frac{d\hat{x}(s; p)}{ds} < 0$ iff $p_0 < p_1$. Clearly $\frac{d\hat{x}(s; p)}{ds} = 0$ if $p_0 = p_1$.

First consider the case $p_1 > p_0$, or the aligned type is more likely to send $m = 1$ than the misaligned type. This implies $\frac{dV_\theta(s)}{ds} > 0$, and by FOSD of s in m , we have $\Delta_\theta(p_1, p_0) = E[V_\theta(s)|m = 1] - E[V_\theta(s)|m = 0] > 0$. This is a contradiction as both types strictly prefer $m = 1$, implying $p_1 = p_0 = 1$. Similarly, if $p_1 < p_0$, we have $\frac{dV_\theta(s)}{ds} < 0$, such that $\Delta_\theta(p_1, p_0) = E[V_\theta(s)|m = 1] - E[V_\theta(s)|m = 0] < 0$, which gives $p_1 = p_0 = 0$, again a contradiction. Thus it must be the case that $p_1 = p_0$. This is an equilibrium strategy because $V_\theta(s)$ is constant in s , hence $\Delta_\theta(p_1, p_0) = V_\theta(1 - 1) = 0$, as we are integrating two densities over their entire support. Hence the equilibrium reporting strategy is $p_1 = p_0 = p^* \in [0, 1]$. There is a continuum of pooling equilibria in which both types choose $m = 1$ with the same probability, making the pledge m uninformative. Thus $\hat{x}$ collapses to that of the baseline model, and optimal effort is equivalent across models. $\square$

I first prove a series of lemmas prior to proving Proposition 4.

Lemma A.1. *If $\rho > 0$, $p_1^* = p_0^* = 1$ is the only pooling equilibrium. If $\rho < 0$, $p_1^* = p_0^* = 0$ is the only pooling equilibrium. There exists a $\bar{\rho}$ such that if $|\rho| > \bar{\rho}$ then these pooling equilibria are the only equilibria of the game.*

Proof of Lemma A.1. Let $\rho > 0$. Leader-type θ chooses $m = 1$ iff $\Delta_\theta(p_1, p_0) + \rho > 0$. From Proposition 3, if $p_1 = p_0$, then $\Delta_\theta(p_1, p_0) = 0$ so any strategy where $p_1 = p_0$ yields the payoff $\rho > 0$. Clearly only $p_1 = p_0 = 1$ satisfies this condition. A symmetric argument can be made for $\rho < 0$.

Now observe that $V_\theta(s)$ is bounded below by zero as a leader can always choose $a_\theta = 0$ and is bounded above by Ψ . Note also that $\phi(\sqrt{\tau}(s-1)) > \phi(\sqrt{\tau}s)$ iff $s > \frac{1}{2}$ such that, by definition of $\Delta_\theta(p_1, p_0)$,

$$\begin{aligned}\Delta_\theta(p_1, p_0) &\leq \Psi \int_{\frac{1}{2}}^{\infty} \sqrt{\tau} \left(\phi(\sqrt{\tau}(s-1)) - \phi(\sqrt{\tau}s) \right) ds \\ &= \Psi \left(2\Phi\left(\frac{\sqrt{\tau}}{2}\right) - 1 \right) \equiv \bar{\rho}.\end{aligned}$$

Thus if $\rho > \bar{\rho}$ then $m = 1$ is a dominant strategy for any value of $\Delta_\theta(p_1, p_0)$ and if $\rho < -\bar{\rho}$ then $m = 0$ is a dominant strategy for any value of $\Delta_\theta(p_1, p_0)$. $\square$

Lemma A.2. *Full separation is not an equilibrium. Any informative equilibrium requires $|\rho| < \bar{\rho}$ and for both types to mix, $p_\theta \in (0, 1)$.*

Proof of Lemma A.2. Recall that $\hat{m}_\theta = \hat{p}_\theta \phi(\sqrt{\tau}(s-1)) + (1 - \hat{p}_\theta) \phi(\sqrt{\tau}s)$ is the total probability that the IO's report is realized as the value s given voter's conjectures. The likelihood ratio of the IO's report s is then $\ell(s) = \frac{\hat{m}_0}{\hat{m}_1}$ where $\ell(s) > 1$ means the report is more likely to signal misalignment and $\ell(s) < 1$ means the report is more likely to signal alignment. Recall also that the voter's cutoff in equilibrium is $\hat{x} = \frac{a_1 + a_0}{2} + \frac{\log \ell(s)}{\zeta(a_1 - a_0)}$, and the leader's first-order condition for optimal effort satisfies $\Psi \sqrt{\zeta} \phi(\sqrt{\zeta}(a_\theta - \hat{x})) = \lambda_\theta a_\theta$. Now define $\eta_\theta = \sqrt{\zeta}(a_\theta - \hat{x})$. Using the definition of η_θ and the voter's cutoff $\hat{x}$, it follows that

$$\begin{aligned}\eta_1^2 - \eta_0^2 &= \zeta \left((a_1 - \hat{x})^2 - (a_0 - \hat{x})^2 \right) \\ \eta_1^2 - \eta_0^2 &= -2 \log \ell(s) \\ -\frac{\eta_1^2 - \eta_0^2}{2} &= \log \ell(s) \\ \frac{\phi(\eta_1)}{\phi(\eta_0)} &= \ell(s).\end{aligned}$$

Now dividing the leader's first-order conditions gives

$$\frac{\lambda_1 a_1}{\lambda_0 a_0} = \ell(s).$$

Since in equilibrium $a_1 > a_0$, we know that $\ell(s) > \frac{\lambda_1}{\lambda_0} > 0$. Now suppose that $p_1 = 1$ and p_0 , which is a fully separating strategy in which the aligned type always sends $m = 1$ and the misaligned type always sends $m = 0$. In this case, $\ell(s) = \frac{\phi(\sqrt{\tau}s)}{\phi(\sqrt{\tau}(s-1))} = \exp(\tau(\frac{1}{2} - s))$. Now for any $s > \frac{1}{2} + \frac{1}{\tau} \log \frac{\lambda_0}{\lambda_1}$ we have $\ell(s) < \frac{\lambda_1}{\lambda_0}$, contradicting the equilibrium condition. Hence

$p_1 = 1, p_0 = 0$ is not an equilibrium. A symmetric argument can be made ruling out $p_1 = 0, p_0 = 1$.

Now consider semi-separation, which can only happen if $|\rho| < \bar{\rho}$ per Lemma A.1. Recall from the proof of Proposition 1 that the second-order condition for equilibrium requires concavity, or $\zeta < \frac{\lambda_\theta \sqrt{2\pi e}}{\Psi} \Leftrightarrow \frac{\Psi \zeta}{\sqrt{2\pi e}} < \lambda_1$. Define $\kappa = \frac{\Psi \zeta}{\sqrt{2\pi e}}$. The first-order condition's slope, or the marginal electoral return from effort, is thus bounded by κ . Applying this between a_0 and a_1 and then substituting in the first-order condition gives

$$-\kappa(a_1 - a_0) \leq \lambda_1 a_1 - \lambda_0 a_0 \leq \kappa(a_1 - a_0)$$

$$\begin{aligned} \frac{\lambda_0 + \kappa}{\lambda_1 + \kappa} &\leq \frac{a_1}{a_0} \leq \frac{\lambda_0 - \kappa}{\lambda_1 - \kappa} \\ \frac{\lambda_1(\lambda_0 + \kappa)}{\lambda_0(\lambda_1 + \kappa)} &\leq \ell(s) \leq \frac{\lambda_1(\lambda_0 - \kappa)}{\lambda_0(\lambda_1 - \kappa)}, \end{aligned}$$

where the last line follows from multiplying by $\frac{\lambda_1}{\lambda_0}$ and using the equilibrium requirement on $\ell(s)$. Rewrite $\ell(s) = \frac{p_0 r(s) + 1 - p_0}{p_1 r(s) + 1 - p_1}$, where $r(s) = \frac{\phi(\sqrt{\tau}(s-1))}{\phi(\sqrt{\tau}s)}$. Then as $r(s)$ becomes arbitrarily large or arbitrarily small, we have

$$\begin{aligned} \frac{\lambda_1(\lambda_0 + \kappa)}{\lambda_0(\lambda_1 + \kappa)} p_1 &\leq p_0 \leq \frac{\lambda_1(\lambda_0 - \kappa)}{\lambda_0(\lambda_1 - \kappa)} p_1 \\ \frac{\lambda_1(\lambda_0 + \kappa)}{\lambda_0(\lambda_1 + \kappa)} (1 - p_1) &\leq 1 - p_0 \leq \frac{\lambda_1(\lambda_0 - \kappa)}{\lambda_0(\lambda_1 - \kappa)} (1 - p_1). \end{aligned}$$

If $p_1 = 0$, the first line forces $p_0 = 0$. If $p_1 = 1$, the second line forces $p_0 = 1$. Because $\frac{\lambda_1(\lambda_0 + \kappa)}{\lambda_0(\lambda_1 + \kappa)} > 0$, if the misaligned type chooses $p_0 = 1$ or $p_0 = 0$ then we must also have $p_1 = 1$ or $p_1 = 0$ by the aligned type. Thus any informative equilibrium must have $0 < p_1, p_0 < 1$ where $p_1 \neq p_0$. Specifically, these probabilities must satisfy $\Delta_1(p_1, p_0) = \Delta_0(p_1, p_0) = -\rho$. $\square$

Lemma A.3. *Let $\tilde{a}_\theta$ be optimal effort in the baseline model without the IO. There is a weight $\omega \in (0, 1)$ such that every equilibrium effort solution satisfies*

$$\frac{a_1(s)}{\tilde{a}_1} \leq 1 - \omega + \omega \ell(s).$$

Proof of Lemma A.3. Define $b_\theta = \sqrt{\zeta} a_\theta$, $y = \sqrt{\zeta} \hat{x}$, and $\alpha_\theta = \Psi \zeta / \lambda_\theta$. The first-order conditions for optimal effort can be expressed as

$$b_\theta = \alpha_\theta \phi(\eta_\theta),$$

where $\eta_\theta = b_\theta - y$. By concavity from the second-order condition, we know $D_\theta \equiv 1 + \eta_\theta b_\theta > 0$.

For $t_\theta(y) = 1/\phi(\eta_\theta(y))$, differentiation gives

$$t'_\theta = -\frac{\eta_\theta t_\theta}{D_\theta}.$$

$$t''_\theta = \frac{t_\theta}{D_\theta^3} (1 + \eta_\theta^2 + 2b_\theta \eta_\theta^3).$$

Now choose

$$\omega = \frac{D_0^B}{D_1^B + D_0^B} = \frac{1 - vb_0^B}{2(1 + v^2)} \in (0, 1)$$

and define

$$T(y) = (1 - \omega)t_1(y) + \omega t_0(y),$$

where $\tilde{y} = (\tilde{b}_1 + \tilde{b}_0)/2$ is the voter's cutoff in the baseline model (where tilde denotes baseline effort), and let $v = (\tilde{b}_1 - \tilde{b}_0)/2 > 0$. Then $\tilde{\eta}_1 = v$, $\tilde{\eta}_0 = -v$. By construction, $T'(\tilde{y}) = 0$. We show that $\tilde{y}$ uniquely minimizes T . Below $I = [\alpha_0\phi(0), \alpha_1\phi(0)]$, both $\eta_\theta > 0$, so $T' < 0$; above I , both $\eta_\theta < 0$, so $T' > 0$. On I , $\eta_1 \geq 0 \geq \eta_0$, immediately giving $t'_1 > 0$. For the misaligned type, put $u = -\eta_0 \geq 0$. Concavity implies

$$u + b_0 = y \leq \alpha_1\phi(0) < \sqrt{e},$$

so $4b_0u < e$ and $u^2 < e$. Consequently,

$$1 + u^2 - 2b_0u^3 \geq 1 + (1 - e/2)u^2 > 1 + e - e^2/2 > 0.$$

Thus $t''_0 > 0$ and $T'' > 0$ on I . Since $\tilde{y} \in I$ and $T'(\tilde{y}) = 0$, it is the unique global minimum. Therefore,

$$\frac{1 - \omega}{\phi(\eta_1)} + \frac{\omega}{\phi(\eta_0)} \geq T(\tilde{y}) = \frac{1}{\phi(v)}.$$

Multiplying by $\phi(\eta_1)$, and using the first-order conditions and the equilibrium identity $\phi(\eta_1)/\phi(\eta_0) = \ell(s)$, yields

$$\frac{a_1(s)}{\tilde{a}_1} \leq 1 - \omega + \omega\ell(s),$$

as desired. □

Lemma A.4. *Let $\tilde{a}_\theta$ be optimal effort in the baseline model without the IO and $a_\theta(s)$ be optimal effort in any informative (semi-separating) equilibrium. In any informative equilibrium,*

$$\frac{E[a_1(s)]}{\tilde{a}_1} = \frac{E[a_0(s)]}{\tilde{a}_0} < 1.$$

Proof of Lemma A.4. By Lemma A.3, we know

$$\frac{a_1(s)}{\tilde{a}_1} \leq 1 - \omega + \omega \ell(s).$$

Multiplying by $\sqrt{\tau}m_1 = p_1\sqrt{\tau}\phi(\sqrt{\tau}(s-1)) + (1-p_1)\sqrt{\tau}\phi(\sqrt{\tau}s)$, the density of the IO report s in equilibrium, and integrating with respect to s gives

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{a_1(s)}{\tilde{a}_1} \sqrt{\tau}m_1(s)ds &\leq (1-\omega) \int_{-\infty}^{\infty} \sqrt{\tau}m_1(s)ds + \omega \int_{-\infty}^{\infty} \ell(s)\sqrt{\tau}m_1(s)ds \\ \frac{E[a_1]}{\tilde{a}_1} &\leq (1-\omega) + \omega \int_{-\infty}^{\infty} \sqrt{\tau}m_0(s)ds \\ \frac{E[a_1]}{\tilde{a}_1} &\leq 1 - \omega + \omega = 1. \end{aligned}$$

The second line exploits $\ell(s) = \frac{m_0(s)}{m_1(s)}$ from Lemma A.2 and the third line uses that the signal densities integrate to 1. The inequality is strict because if $p_1 \neq p_0$, then $m_1(s) - m_0(s) = (p_1 - p_0)(\phi(\sqrt{\tau}(s-1)) - \phi(\sqrt{\tau}s))$ which is nonzero except at $s = \frac{1}{2}$. Hence $\ell(s) \neq 1$ almost surely and the bound is strict almost surely. It is straightforward that $\frac{E[a_1]}{\tilde{a}_1} = \frac{E[a_0]}{\tilde{a}_0}$ given the first-order conditions. $\square$

Proof of Proposition 4. Follows from Lemmas A.1-A.4. $\square$

Consider a two-period extension of the model. The incumbent's type remains fixed across periods, while the reporting and policy shocks are independent over time (i.e., ν_t and ε_t are i.i.d. draws). Let $h_1 = (s_1, x_1)$ be the history that contains the IO's first period report and the first period governance outcome. If the incumbent is retained after the first period, the voter enters the second period with belief $q_2(h_1) = P(\theta = 1|h_1)$, and a potential challenger is aligned with probability γ . For variables with two subscripts, let the first subscript denote leader type and the second denote time (e.g., $\hat{a}_{12}$ is the voter's conjecture about the aligned type's effort in the second period).

Proposition A.1. *In a two-period version of the model, the voter reelects a previously retained incumbent if and only if*

$$x_2 \geq \hat{x}_2(h_1, s_2),$$

where

$$\hat{x}_2(h_1, s_2) = \frac{\hat{a}_{12}(h_1, s_2) + \hat{a}_{02}(h_1, s_2)}{2} + \frac{\log\left(\frac{\gamma}{1-\gamma}\right) - \log\left(\frac{q_2(h_1)}{1-q_2(h_1)}\right) + \log\left(\frac{\hat{m}_{02}(s_2 | h_1)}{\hat{m}_{12}(s_2 | h_1)}\right)}{\zeta(\hat{a}_{12}(h_1, s_2) - \hat{a}_{02}(h_1, s_2))}.$$

Proof of Proposition A.1. Define the voter's average conjectured effort and the conjectured effort difference in the first period as $\bar{a}_1(s_1) = \frac{\hat{a}_{11}(s_1) + \hat{a}_{01}(s_1)}{2}$ and $d_1(s_1) = \hat{a}_{11}(s_1) - \hat{a}_{01}(s_1) > 0$. The corresponding second-period quantities are $\bar{a}_2(h_1, s_2) = \frac{\hat{a}_{12}(h_1, s_2) + \hat{a}_{02}(h_1, s_2)}{2}$ and $d_2(h_1, s_2) = \hat{a}_{12}(h_1, s_2) - \hat{a}_{02}(h_1, s_2) > 0$.

Observe that this means the voter's retention threshold in the first period can be written as $\hat{x}_1(s_1) = \bar{a}_1(s_1) + \frac{\log\left(\frac{\hat{m}_{01}(s_1)}{\hat{m}_{11}(s_1)}\right)}{\zeta d_1(s_1)}$.

After observing $h_1 = (s_1, x_1)$, the voter's posterior odds that the incumbent is aligned are

$$\frac{q_2(h_1)}{1 - q_2(h_1)} = \frac{\gamma}{1 - \gamma} \frac{\hat{m}_{11}(s_1)}{\hat{m}_{01}(s_1)} \frac{\phi(\sqrt{\zeta}(x_1 - \hat{a}_{11}(s_1)))}{\phi(\sqrt{\zeta}(x_1 - \hat{a}_{01}(s_1)))}.$$

Since x_1 is normally distributed, $\log \frac{\phi(\sqrt{\zeta}(x_1 - \hat{a}_{11}(s_1)))}{\phi(\sqrt{\zeta}(x_1 - \hat{a}_{01}(s_1)))} = \zeta d_1(s_1)(x_1 - \bar{a}_1(s_1))$, so the log posterior odds are

$$\log \left(\frac{q_2(h_1)}{1 - q_2(h_1)} \right) = \log \left(\frac{\gamma}{1 - \gamma} \right) + \log \left(\frac{\hat{m}_{11}(s_1)}{\hat{m}_{01}(s_1)} \right) + \zeta d_1(s_1)(x_1 - \bar{a}_1(s_1)).$$

Substituting the cutoff from the first period into this expression yields

$$\log \left(\frac{q_2(h_1)}{1 - q_2(h_1)} \right) - \log \left(\frac{\gamma}{1 - \gamma} \right) = \zeta d_1(s_1)(x_1 - \hat{x}_1(s_1)).$$

Now consider the voter's second period decision. After observing s_2 and x_2 , the voter's posterior odds that the retained incumbent is aligned are

$$\frac{\mu_2(x_2, s_2 | h_1)}{1 - \mu_2(x_2, s_2 | h_1)} = \frac{q_2(h_1)}{1 - q_2(h_1)} \frac{\hat{m}_{12}(s_2 | h_1)}{\hat{m}_{02}(s_2 | h_1)} \frac{\phi(\sqrt{\zeta}(x_2 - \hat{a}_{12}(h_1, s_2)))}{\phi(\sqrt{\zeta}(x_2 - \hat{a}_{02}(h_1, s_2)))}.$$

The voter retains the incumbent if and only if the posterior probability that she is aligned is at least the probability that a new challenger is aligned, $\mu_2(x_2, s_2 | h_1) \geq \gamma$. In odds form, this condition is

$$\frac{q_2(h_1)}{1 - q_2(h_1)} \frac{\hat{m}_{12}(s_2 | h_1)}{\hat{m}_{02}(s_2 | h_1)} \frac{\phi(\sqrt{\zeta}(x_2 - \hat{a}_{12}(h_1, s_2)))}{\phi(\sqrt{\zeta}(x_2 - \hat{a}_{02}(h_1, s_2)))} \geq \frac{\gamma}{1 - \gamma}.$$

As in the first period, the second period outcome likelihood ratio satisfies $\log \frac{\phi(\sqrt{\zeta}(x_2 - \hat{a}_{12}(h_1, s_2)))}{\phi(\sqrt{\zeta}(x_2 - \hat{a}_{02}(h_1, s_2)))} = \zeta d_2(h_1, s_2)(x_2 - \bar{a}_2(h_1, s_2))$, so the log of the retention condition is

$$\log \left(\frac{q_2(h_1)}{1 - q_2(h_1)} \right) + \log \left(\frac{\hat{m}_{12}(s_2 | h_1)}{\hat{m}_{02}(s_2 | h_1)} \right) + \zeta d_2(h_1, s_2)(x_2 - \bar{a}_2(h_1, s_2)) \geq \log \left(\frac{\gamma}{1 - \gamma} \right).$$

Solving for x_2 yields

$$x_2 \geq \bar{a}_2(h_1, s_2) + \frac{\log\left(\frac{\gamma}{1-\gamma}\right) - \log\left(\frac{q_2(h_1)}{1-q_2(h_1)}\right) + \log\left(\frac{\hat{m}_{02}(s_2|h_1)}{\hat{m}_{12}(s_2|h_1)}\right)}{\zeta d_2(h_1, s_2)} \equiv \hat{x}_2(h_1, s_2),$$

which proves the cutoff stated in the proposition.

Finally, substituting for $\log\left(\frac{\gamma}{1-\gamma}\right) - \log\left(\frac{q_2(h_1)}{1-q_2(h_1)}\right)$ gives the equivalent expression

$$\hat{x}_2(h_1, s_2) = \bar{a}_2(h_1, s_2) + \frac{\log\left(\frac{\hat{m}_{02}(s_2|h_1)}{\hat{m}_{12}(s_2|h_1)}\right)}{\zeta d_2(h_1, s_2)} - \frac{d_1(s_1)}{d_2(h_1, s_2)} (x_1 - \hat{x}_1(s_1)).$$

Because the incumbent was retained after the first period, $x_1 \geq \hat{x}_1(s_1)$, or, equivalently, $q_2(h_1) \geq \gamma$. Holding second period conjectures fixed, the reputation inherited from the first period therefore weakly lowers the policy outcome required for reelection in the second period. $\square$